

WEEK-1 SQL ASSIGNMENT

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Tables Creation:

```
create table client_master (  
    clientno varchar(6) primary key check (clientno like 'C%'),  
    name varchar(20) not null,  
    address1 varchar(30),  
    address2 varchar(30),  
    city varchar(15),  
    pincode numeric(8),  
    state varchar(15),  
    baldue numeric(10,2)  
);  
  
create table product_master (  
    productno varchar(6) primary key check (productno like 'P%'),  
    description varchar(15) not null,  
    profitperc numeric(4,2) not null,  
    unitmeasure varchar(10) not null,  
    qtyonhand numeric(8) not null,  
    reorderlvl numeric(8) not null,  
    sellprice numeric(8,2) not null check (sellprice <> 0),  
    costprice numeric(8,2) not null check (costprice <> 0)  
);  
  
create table salesman_master (  
    salesmanno varchar(6) primary key check (salesmanno like 'S%'),  
    salesmanname varchar(20) not null,  
    address1 varchar(30) not null,  
    address2 varchar(30),  
    city varchar(20),  
    pincode numeric(8),  
    state varchar(20),  
    salamt numeric(8,2) not null check (salamt <> 0),  
    tgttoget numeric(6,2) not null,  
    ytdsales numeric(6,2) not null,  
    remarks varchar(60)  
);  
  
create table sales_order (  
    orderno varchar(6) primary key check (orderno like 'O%'),  
    clientno varchar(6) references client_master(clientno),  
    orderdate date,  
    delyaddr varchar(25),  
    salesmanno varchar(6) references salesman_master(salesmanno),  
    delytype char(1) check (delytype in ('P','F')),  
    billedyn char(1) check (billedyn in ('Y','N')),  
    delydate date,  
    orderstatus varchar(10) check (orderstatus in ('in  
process','fulfilled','backorder','cancelled'))  
);  
  
create table sales_order_details (  
    orderno varchar(6) references sales_order(orderno),  
    productno varchar(6) references product_master(productno),  
    qtyordered numeric(8),
```

```

        qtydisp numeric(8),
        productrate numeric(10,2),
        constraint pk_sales_order_details primary key (orderno, productno)
    );

```

Insertion of Data:

```

INSERT INTO client_master VALUES
('C0001', 'Rahul Sharma', 'MG Road', 'Sector
5', 'Mumbai', 400001, 'Maharashtra', 1200.50),
('C0002', 'Anita Verma', 'Park Street', NULL, 'Kolkata', 700016, 'West
Bengal', 800.00),
('C0003', 'Suresh Reddy', 'Ameerpet', 'Near
Metro', 'Hyderabad', 500016, 'Telangana', 1500.75),
('C0004', 'Priya Singh', 'Anna Nagar', NULL, 'Chennai', 600040, 'Tamil
Nadu', 500.00),
('C0005', 'Amit Kumar', 'FC Road', 'Shivaji
Nagar', 'Pune', 411004, 'Maharashtra', 0.00);

```

```

insert into client_master values('C0006', 'tarun Kumar', 'Nizampet
Road', 'Balaji Nagar', 'Bachpally', 411004, 'Hyderabad', 0.00)
INSERT INTO product_master VALUES
('P0001', '1.44 Drive', 15.50, 'Nos', 100, 20, 1200.50, 1000.00),
('P0002', 'Keyboard', 10.00, 'Nos', 200, 30, 800.00, 650.00),
('P0003', 'Mouse', 12.00, 'Nos', 300, 50, 400.00, 300.00),
('P0004', 'Monitor', 18.00, 'Nos', 80, 15, 4500.00, 3800.00),
('P0005', 'Printer', 20.00, 'Nos', 40, 10, 6000.00, 5000.00);

```

```

INSERT INTO salesman_master VALUES
('S0001', 'Ramesh Kumar', 'MG
Road', NULL, 'Mumbai', 400001, 'Maharashtra', 25000.00, 8000.00, 6000.00, 'Good
performer'),
('S0002', 'Sita Rao', 'Brigade
Road', NULL, 'Bangalore', 560001, 'Karnataka', 22000.00, 7500.00, 5800.00, 'Consis
tent'),
('S0003', 'Arjun Patel', 'CG
Road', 'Navrangpura', 'Ahmedabad', 380009, 'Gujarat', 20000.00, 7000.00, 5200.00,
'Average'),
('S0004', 'Meena Iyer', 'T Nagar', NULL, 'Chennai', 600017, 'Tamil
Nadu', 23000.00, 8500.00, 7900.00, 'Excellent'),
('S0005', 'Vikram Singh', 'Connaught
Place', NULL, 'Delhi', 110001, 'Delhi', 26000.00, 9000.00, 8800.00, 'Top seller');

```

```

INSERT INTO sales_order VALUES
('00001', 'C0001', '2025-01-10', 'Mumbai', 'S0001', 'P', 'Y', '2025-01-15', 'fulfi
lled'),
('00002', 'C0002', '2025-01-12', 'Kolkata', 'S0002', 'F', 'Y', '2025-01-18', 'in
process'),
('00003', 'C0003', '2025-01-14', 'Hyderabad', 'S0003', 'P', 'N', '2025-01-20', 'ba
ckorder'),
('00004', 'C0004', '2025-01-16', 'Chennai', 'S0004', 'F', 'Y', '2025-01-22', 'fulf
illed'),
('00005', 'C0005', '2025-01-18', 'Pune', 'S0005', 'P', 'N', '2025-01-25', 'cancell
ed');

```

```

INSERT INTO sales_order_details VALUES
('00001', 'P0001', 10, 10, 1200.50),
('00001', 'P0002', 5, 5, 800.00),

```

```
( '00002' , 'P0003' , 15, 10, 400.00),
( '00003' , 'P0004' , 3, 0, 4500.00),
( '00004' , 'P0005' , 2, 2, 6000.00);
```

Answer the following queries with the help of above schema

1.Display the names of all the clients.

```
select name from client_master;
```

	name
1	Rahul Sharma
2	Anita Verma
3	Suresh Reddy
4	Priya Singh
5	Amit Kumar

2.Display all the clients who are located in Mumbai.

```
select * from client_master where city='Mumbai';
```

	clientno	name	address1	address2	city	pincode	state	baldue
1	C0001	Rahul Sharma	MG Road	Sector 5	Mumbai	400001	Maharashtra	1200.50

3.Display all the products whose selling price is > 2000 and < 5000.

```
select * from product_master where sellprice>2000 and sellprice<5000;
```

	productno	description	profitperc	unitmeasure	qtyonhand	reorderlvl	sellprice	costprice
1	P0004	Monitor	18.00	Nos	80	15	4500.00	3800.00

4.Display Name, City and State of Clients not in the state of Maharashtra.

```
select name,city,state from client_master where state not in
('Maharashtra');
```

	name	city	state
1	Anita Verma	Kolkata	West Bengal
2	Suresh Reddy	Hyderabad	Telangana
3	Priya Singh	Chennai	Tamil Nadu

5.Display all the information of client_no C0001 and C0002.

```
select * from client_master where clientno in ('C0001', 'C0002');
```

	clientno	name	address1	address2	city	pincode	state	baldue
1	C0001	Rahul Sharma	MG Road	Sector 5	Mumbai	400001	Maharashtra	1200.50
2	C0002	Anita Verma	Park Street	NULL	Kolkata	700016	West Bengal	800.00

6.Change the selling price of '1.44 drive' to Rs. 1150.50.

```
update product_master set sellprice=1150.50 where description='1.44 Drive';
```

7.Delete the record of client_no C0005.

```
delete from client_master where clientno='C0005';
```

8.Display the clients who stay in a city whose second letter is 'a'.

```
select * from client_master where city like '_a%';
```

	clientno	name	address1	address2	city	pincode	state	baldue
1	C0006	tarun Kumar	Nizampet Road	Balaji Nagar	Bachpally	411004	Hyderabad	0.00

9.Count the number of products having price greater than or equal to 1500.

```
select count(*) as product_count from product_master where sellprice>=1500;
```

	product_count
1	2

10.Display qtyordered, qtydisp and balancedqty (not in table).

```
select qtyordered, qtydisp, (qtyordered - qtydisp) as balancedqty from sales_order_details;
```

	qtyordered	qtydisp	balancedqty
1	10	10	0
2	5	5	0
3	15	10	5
4	3	0	3
5	2	2	0

Write Commands to do following

1.Make Client_no as primary key in client_master.

```
alter table client_master
```

```
add constraint pk_clientno primary key (clientno);
```

2.Add a new column phone no in the client_master table.

```
alter table client_master add phoneno varchar(15);
```

3.Add the not null constraint in the product_master table with the column description, profit percent, sell price and cost price.

```
alter table product_master
alter column description varchar(15) not null;

alter table product_master
alter column profitperc numeric(4,2) not null;

alter table product_master
alter column sellprice numeric(8,2) not null;

alter table product_master

alter column costprice numeric(8,2) not null;
```

4.Change size of name column to 60 in client_master table.

```
alter table client_master

alter column name varchar(60);
```

5.Remove pincode column from table.

```
alter table client_master

drop column pincode;
```

Define in 1 or 2 lines and give one example also

1.Recursive Relationship

A Recursive Relationship occurs when a table is related to itself.

Example: Employee table where each employee has a manager_id that refers to the another employee in the same table.

employee(emp_id, name, manager_id)

2.Composite key.

It is a Primary key made up of two or more columns to uniquely identify a record

Example: In an order table order_number ,product_number together identify the records uniquely

Primary key (order_number,product_number);

3.The 'like' operator with pattern matching.

It is used to search for a specified pattern in a column

Example:Find names of clients whose name starts with 'a'

```
select name from client_master where name like 'a%';
```

	name
1	Anita Verma

4.Drop Table command.

It delete the structure and also the data of the table permanently

Example: delete client_master table

```
drop table client_master
```

5.Full Outer Join.

A Full Outer Join returns all records from the both tables.if match exists it shows NULL.

Example:

```
Select * from client_master c full outer join sales_order s on  
c.clientno=s.clientno;
```

	clientno	name	address1	address2	city	state	baldue	phoneno	orderno	clientno	orderno	delyaddr	salesmanno	delytype	billedyn	delydate	orderstatus
1	C0001	Rahul Sharma	MG Road	Sector 5	Mumbai	Maharashtra	1200.50	NULL	O0001	C0001	2025-01-10	Mumbai	S0001	P	Y	2025-01-15	fulfilled
2	C0001	Rahul Sharma	MG Road	Sector 5	Mumbai	Maharashtra	1200.50	NULL	O0007	C0001	2026-01-03	Mumbai	S0001	P	N	2026-01-03	in process
3	C0001	Rahul Sharma	MG Road	Sector 5	Mumbai	Maharashtra	1200.50	NULL	O0008	C0001	2026-01-04	Mumbai	S0001	P	Y	2026-01-04	fulfilled
4	C0002	Anita Verma	Park Street	NULL	Kolkata	West Bengal	800.00	NULL	O0002	C0002	2025-01-12	Kolkata	S0002	F	Y	2025-01-18	in process
5	C0002	Anita Verma	Park Street	NULL	Kolkata	West Bengal	800.00	NULL	O0009	C0002	2026-01-04	Kolkata	S0002	F	Y	2026-01-04	fulfilled
6	C0003	Suresh Reddy	Armeerpeth	Near Metro	Hyderabad	Telangana	1500.75	NULL	O0003	C0003	2025-01-14	Hyderabad	S0003	P	N	2025-01-20	backorder
7	C0004	Priya Singh	Anna Nagar	NULL	Chennai	Tamil Nadu	500.00	NULL	O0004	C0004	2025-01-16	Chennai	S0004	F	Y	2025-01-22	fulfilled
8	C0006	tarun Kumar	Nizampet Road	Balaji Nagar	Bachpally	Hyderabad	0.00	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Write queries for following descriptions: (Joins)

1.Find out the products, which have been sold to 'Ivan Bayross'.

```
select p.productno,p.description from client_master c  
join sales_order s  
on c.clientno=s.clientno  
join sales_order_details sd  
on s.orderno=sd.orderno  
join product_master p  
on sd.productno=p.productno
```

```
where c.name='Ivan Bayross';
```

	productno	description
1	P00006	T-Shirts

2.Finding out the products and their quantities that will have to be delivered in the current month.

```
select p.productno,p.description,(sd.qtyordered-sd.qtydisp) as  
qty_to_be_delivered  
from sales_order s  
join sales_order_details sd  
on s.orderno=sd.orderno  
join product_master p  
on sd.productno=p.productno  
where month(s.delydate)=month(getdate())
```

```
and year(s.delydate)=year(getdate());
```

	productno	description	qty_to_be_delivered
1	P0004	Monitor	6

3.Listing the ProductNo and description of constantly sold (i.e. rapidly moving) products.

```
select p.productno,p.description from product_master p  
join sales_order_details sd
```

```

on p.productno=sd.productno
group by p.productno,p.description

having count(sd.orderno)>1;

```

	productno	description
1	P0004	Monitor

4.Finding the names of clients who have purchased 'Trousers'.

```

select c.name from client_master c
join sales_order s
on c.clientno=s.clientno
join sales_order_details sd
on s.orderno=sd.orderno
join product_master p
on sd.productno=p.productno

where p.description='Trousers';

```

	name
1	Rahul Sharma

5.Listing the products and orders from customers who have ordered less than 5 units of 'Pull Overs'.

```

select
    s.orderno,
    p.description,
    sd.qtyordered
from sales_order s
join sales_order_details sd
on s.orderno = sd.orderno
join product_master p
on sd.productno = p.productno
where p.description = 'Pull Overs'

and sd.qtyordered < 5;

```

	orderno	description	qtyordered
1	O0009	Pull Overs	3

Write queries for following descriptions: (Subqueries)

1.Finding the non-moving products i.e. products not being sold.

```

select * from product_master where productno
not in (select productno from sales_order_details);

```

	productno	description	profitperc	unitmeasure	qtyonhand	reorderlvl	sellprice	costprice
1	P0009	Calculator	8.00	Piece	50	10	900.00	700.00

2. Finding the name and complete address for the customer who has placed Order number 'O0001'.

```
select name, address1, address2, city, state from client_master
```

```
where clientno=(select clientno from sales_order where orderno='O0001');
```

	name	address1	address2	city	state
1	Rahul Sharma	MG Road	Sector 5	Mumbai	Maharashtra

3. Finding the clients who have placed orders before the month of May'02.

```
select name from client_master where clientno in (select clientno from sales_order where orderdate<'2025-05-01');
```

	name
1	Rahul Sharma
2	Anita Verma
3	Suresh Reddy
4	Priya Singh

Write Commands to do following

1. Display system date as Saturday, February 11, 2012.

```
select FORMAT(getdate(), 'dddd, MMMM dd, yyyy') as SystemDate;
```

	SystemDate
1	Sunday, January 04, 2026

2. Display Balance Due from Client master as \$99,999.99.

```
select format(baldue, '$#,##0.00') as BalanceDue  
from client_master;
```

	BalanceDue
1	\$1,200.50
2	\$800.00
3	\$1,500.75
4	\$500.00
5	\$0.00

3. Display message as 'Salesman Aman sold goods of 50 while given target was 100.'

```
select  
'Salesman ' + salesmanname +  
' sold goods of ' + CAST(ytdsales as varchar) +  
' while given target was ' + CAST(tgttoget as varchar) + '.'  
as Message  
from salesman_master
```

```
where salesmanname = 'Aman';
```

	Message
1	Salesman Aman sold goods of 50.00 while given tar...

4.Display your Age in Years.

```
select datediff(year, '2003-10-05', getdate()) as age;
```

	age
1	23